THREE EQUIVALENT DESCRIPTIONS OF THE QUOTIENT TOPOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The quotient topology on the target of a surjective map, and the notions of the strongest topology making a map continuous. Set up in the author's section on submersive maps, immediately before the counted statement. Fixes the three descriptions whose equivalence the statement asserts.

Lemma 0.1. *Let X be a topological space. Let Y be a set and let $f : X \to Y$ be a surjective map of sets. The quotient topology on Y is the topology characterized by each of the following statements:*

1. *it is the strongest topology on Y such that f is continuous,*
2. *a subset V of Y is open if and only if $f^{-1}(V)$ is open,*
3. *a subset Z of Y is closed if and only if $f^{-1}(Z)$ is closed.*

Proof. The set $\mathcal{T} = \{V \subset Y | f^{-1}(V) \text{ is open}\}$ is a topology on Y. Firstly $\emptyset = f^{-1}(\emptyset)$ and $f^{-1}(Y) = X$. So $\mathcal{T}$ contains $\emptyset$ and Y.

Let $(V_i)_{i \in I}$ be a family of elements $V_i \in \mathcal{T}$. Then

$$\bigcup\nolimits_{i \in I} f^{-1}(V_i) = f^{-1}\left(\bigcup\nolimits_{i \in I} V_i\right)$$

Thus $\bigcup_{i \in I} V_i \in \mathcal{T}$ as $\bigcup_{i \in I} f^{-1}(V_i)$ is open in X. Furthermore if $V_1, V_2 \in \mathcal{T}$ then

$$f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2)$$

So $V_1 \cap V_2 \in \mathcal{T}$ because $f^{-1}(V_1) \cap f^{-1}(V_2)$ is open in X.

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Finally a topology on Y such that f is continuous is included in $\mathcal{T}$ according to the definition of a continuous function, so $\mathcal{T}$ is the strongest topology on Y such that f is continuous. It proves that (1) and (2) are equivalent.

Finally, (2) and (3) equivalence follows from $f^{-1}(X \setminus E) = Y \setminus f^{-1}(E)$ for all subsets $E \subset X$. $\square$